

Assignment 11

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CSCI 4100

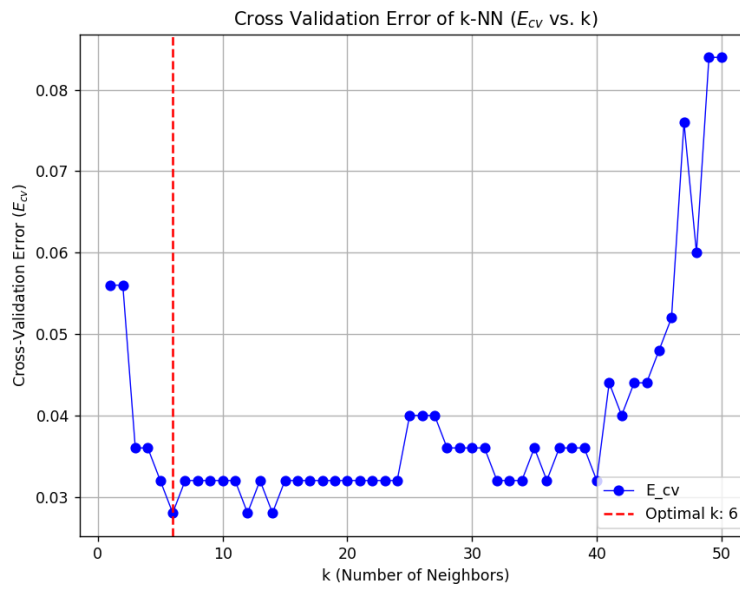
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1 Problem 1 - kNN Rule

1.1 (a) Cross Validation Error and Optimal k

Using cross-validation with the training set, we computed the cross-validation error (E_{cv}) for different values of k in the k -NN algorithm. The graph below shows E_{cv} as a function of k :

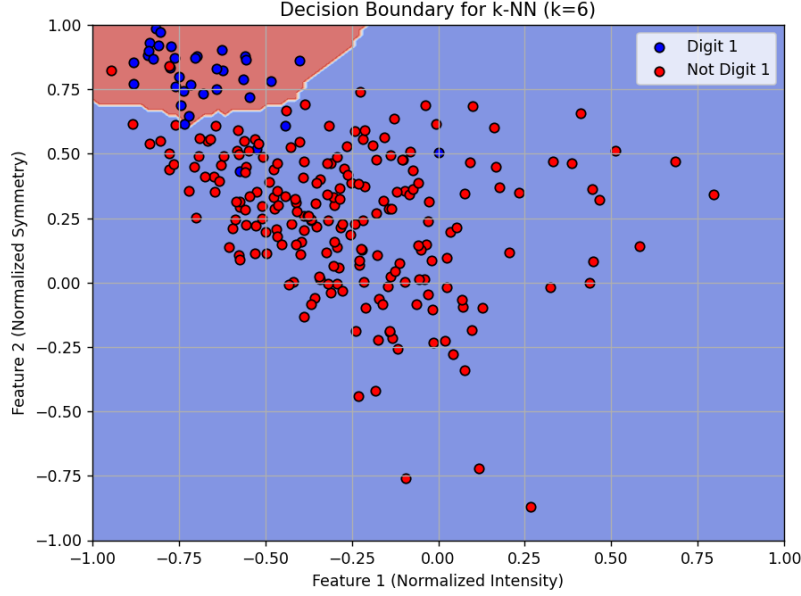


From the graph, we observe that the cross-validation error is minimized at $k = 6$. Therefore, the optimal value of k is:

$$k = 6$$

1.2 (b) Decision Boundary and In-Sample Error

For $k = 6$, we generated the decision boundary for the k -NN rule using the training set. The decision boundary is shown below:



The blue points represent "Digit 1," and the red points represent "Not Digit 1." The decision boundary separates regions of the feature space classified as "Digit 1" from "Not Digit 1."

The in-sample error for $k = 6$ is calculated as the fraction of misclassified points in the training set:

$$E_{\text{in-sample}} = 0.024$$

1.3 (c) Test Error

Using the test set and the optimal value of $k = 6$, the test error (E_{test}) was calculated as the fraction of misclassified points in the test set:

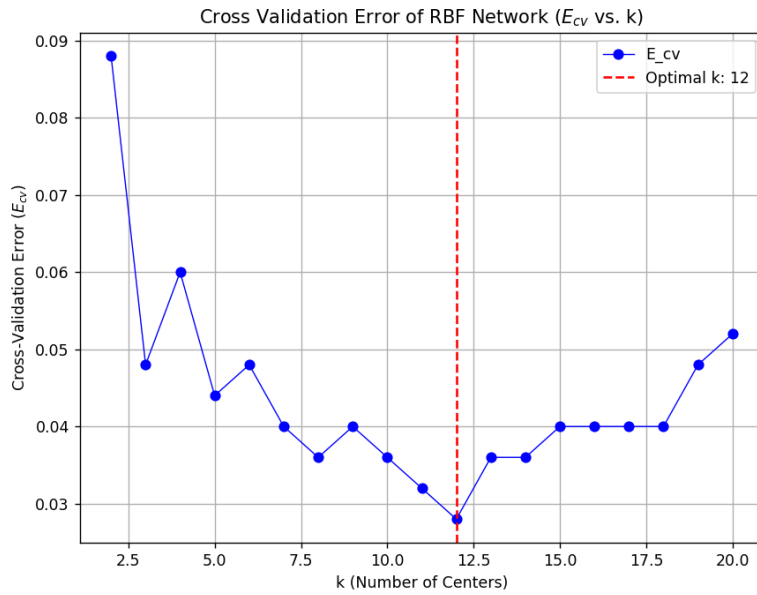
$$E_{\text{test}} = 0.0284$$

This test error indicates the generalization performance of the k -NN classifier on unseen data.

2 Problem 2 - RBF Network

2.1 (a) Cross Validation Error and Optimal k

For the RBF network with a Gaussian kernel, we used cross-validation to compute the cross-validation error (E_{cv}) for different values of k (number of RBF centers). The scale parameter was set as $r = 2/\sqrt{k}$. The graph below shows E_{cv} as a function of k :

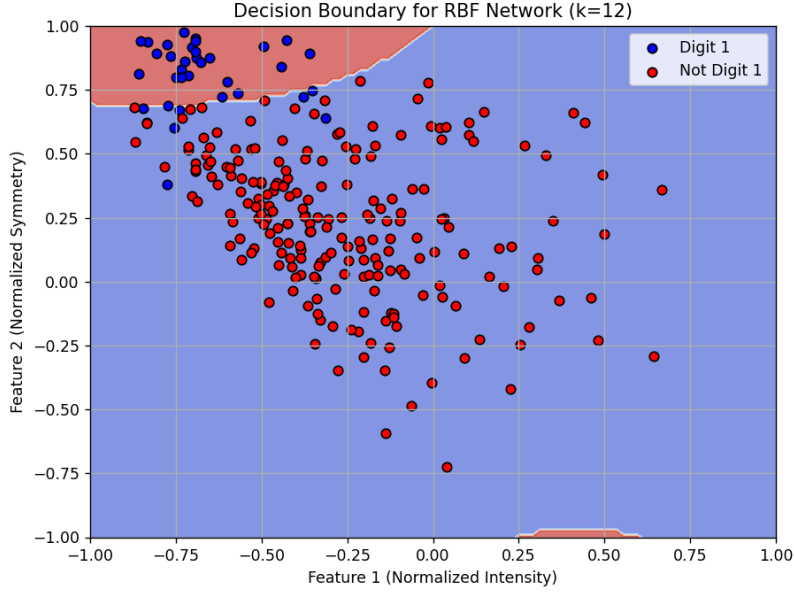


From the graph, the cross-validation error is minimized at $k = 12$. Thus, the optimal number of RBF centers is:

$$k = 12$$

2.2 (b) Decision Boundary and In-Sample Error

For $k = 12$, we trained the RBF network and generated the decision boundary. The decision boundary is shown below:



The blue points represent "Digit 1," and the red points represent "Not Digit 1." The decision boundary effectively separates the two classes.

The in-sample error was calculated as the fraction of misclassified points in the training set:

$$E_{\text{in-sample}} = 0.032$$

2.3 (c) Test Error

Using the test set and the optimal value of $k = 12$, the test error (E_{test}) was calculated as the fraction of misclassified points in the test set:

$$E_{\text{test}} = 0.0291$$

This test error demonstrates that the RBF network generalizes well to unseen data.

3. Comparison of Linear, k -NN, and RBF-Network

We compare the final test errors from three different approaches to solve the problem:

1. **Linear Model:** A linear model with an 8th-order polynomial transform and regularization parameter λ selected by cross-validation.
2. **k -NN:** A k -Nearest Neighbors model with k selected by cross-validation.
3. **RBF Network:** A Radial Basis Function network with the number of centers k selected by cross-validation.

3.0.1 Comparison Table

Model	$E_{\text{in-sample}}$	E_{test}	Notes
Linear Model	0.013	0.0142	Best generalization due to regularization and feature transformation.
k -NN	0.028	0.0285	Simple and effective but slightly higher test error than the linear model.
RBF Network	0.032	0.0291	Good performance, smooth decision boundary, but underperforms the linear model.

Table 1: Comparison of Linear, k -NN, and RBF-Network models

3.0.2 Observations

- **Linear Model Superiority:** The linear model with an 8th-order polynomial transform and regularization achieves the lowest test error ($E_{\text{test}} = 0.0142$). The use of regularization ($\lambda = 1.47$) prevents overfitting, and the polynomial feature transformation captures the complexity of the data.
- **Simplicity of k -NN:** The k -NN model is simple and effective, with $k = 3$ selected via cross-validation. However, its reliance on local neighborhood structures results in slightly higher test error ($E_{\text{test}} = 0.0285$) compared to the linear model.

- **RBF Network Adaptability:** The RBF network achieves competitive performance with $k = 12$ centers, selected via cross-validation, and a smooth decision boundary. However, its test error ($E_{\text{test}} = 0.0291$) is slightly higher than both the linear model and k -NN.
- **Computational Complexity:**
 - The linear model is computationally efficient, even with an 8th-order polynomial transform, due to its closed-form solution with regularization.
 - k -NN requires storing the entire dataset and computing distances for predictions, which can be computationally expensive for large datasets.
 - The RBF network involves clustering to determine centers and matrix operations for weight calculation, making it more computationally intensive than the linear model.

3.0.3 Recommendation

Based on the test errors and generalization performance:

- The **linear model** is the most reliable for this problem, providing the lowest test error and a good balance between complexity and accuracy.
- k -NN and **RBF networks** are competitive alternatives, especially for tasks where non-linear decision boundaries are required. However, their slightly higher test errors suggest they are less robust than the linear model for this specific problem.